

16. Conclusion

The **Conclusion** section summarizes the overall outcome of the entire API testing project executed on Fake Store API using Postman and Newman CLI. It highlights what was achieved, what was validated, and how the testing process improved API quality and reliability.

16.1 Overall Project Summary

In this project, I performed **end-to-end REST API testing** covering all major modules such as:

- Authentication
- Products
- Categories
- Cart
- Users

Each API was tested using:

- Postman (manual + Collection Runner)
 - Newman CLI (automation)
 - HTML reports (result analysis)
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16.2 What was achieved

◆ Functional Validation

- All CRUD operations were successfully validated:
 - Create (POST)
 - Read (GET)
 - Update (PUT/PATCH)
 - Delete (DELETE)
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◆ Data Validation

- Verified response body structure
- Ensured correct JSON format
- Validated required fields like:
 - id

- title
 - price
 - category
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◆ Status Code Validation

- Verified correct HTTP responses:
 - 200 → Successful requests
 - 201 → Resource created
 - 404 → Invalid resource handling
 - 401 → Unauthorized access handling
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◆ Performance Validation

- Response time was within acceptable limits for most APIs
 - No major performance bottlenecks observed
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◆ Automation Execution

- Entire test suite executed using Newman CLI
 - Integrated Postman collection with command-line execution
 - Generated automated test reports
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16.3 Key Learnings

Through this project, I learned:

- How REST APIs work in real-time applications
 - Importance of environment variables in automation
 - How to structure API test collections properly
 - How to integrate Postman with Newman for CI/CD
 - How to debug API failures in real-world scenarios
 - Importance of response validation and schema checks
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16.4 Challenges Overcome

- Fixed base URL configuration issues

- Resolved environment variable export problems
 - Solved Newman invalid URI errors
 - Corrected collection execution dependencies
 - Fixed HTML report generation issues
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16.5 Final Outcome

At the end of this project:

- All APIs were successfully tested and validated
 - Automation framework was stable and reusable
 - Newman execution worked successfully
 - HTML reports provided clear execution insights
 - API system was confirmed as **functionally stable and reliable**
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